



Experimental Optimization and Predictive Modeling of a Tubular Wind Energy Conversion System Using Response Surface Methodology

Nayeemuddin Mohammed^{1*}, Feroz Shaik²

¹ Department of Civil Engineering, Prince Mohammad Bin Fahd University, 31952 Al Khobar, Kingdom of Saudi Arabia

² Department of Mechanical Engineering, Prince Mohammad Bin Fahd University, 31952 Al Khobar, Kingdom of Saudi Arabia

* Correspondence: Nayeemuddin Mohammed (mnayeemuddin@pmu.edu.sa)

Received: 05-16-2026

Revised: 06-30-2026

Accepted: 07-18-2026

Citation: N. Mohammed and F. Shaik, "Experimental optimization and predictive modeling of a tubular wind energy conversion system using response surface methodology," *J. Complex Multiphys. Eng. Syst.*, vol. 1, no. 3, pp. 307–318, 2026. <https://doi.org/10.56578/jcmes010306>.



© 2026 by the author(s). Licensee Acadlore Publishing Services Limited, Hong Kong. This article can be downloaded for free, and reused and quoted with a citation of the original published version, under the CC BY 4.0 license.

Abstract: Small-scale wind energy systems continue to attract attention as distributed renewable energy solutions; however, improving electrical output under varying operating conditions remains a challenge due to nonlinear interactions between environmental and operational parameters. This study investigates the optimization and predictive modelling of the current output of a tubular wind energy conversion system (WECS) through a statistically guided experimental framework. A response surface methodology based on central composite design (RSM–CCD) was employed to evaluate the combined influence of operating time and wind speed on system performance. Experimental observations were analysed using a quadratic polynomial model and analysis of variance (ANOVA) to establish the response relationship and identify operating conditions associated with improved electrical output. The results showed that the developed model achieved strong agreement between predicted and experimental responses with a coefficient of determination (R^2) of 0.962. Wind speed was identified as the dominant factor affecting current generation. The optimization analysis showed that the highest current output of 491.897 A was obtained at an operating time of 30 min and a wind speed of 6.88 m/s, corresponding to a desirability value of 0.998. The results indicate that the proposed modelling framework provides an effective approach for describing system response and identifying favourable operating conditions for compact wind energy systems. This study offers a practical methodology for experimental optimization and supports further development of data-driven performance enhancement strategies in renewable energy applications.

Keywords: Tubular wind energy system; Response surface methodology; Central composite design; Experimental optimization; Predictive modelling; System response analysis

1 Introduction

The increasing global demand for energy and the continued transition toward low-carbon energy systems have intensified interest in renewable power generation technologies. Among available alternatives, wind energy remains one of the most mature and scalable solutions due to its ability to convert naturally available kinetic energy into usable electrical output. Wind energy conversion systems (WECS) have therefore become increasingly important in both centralized and distributed energy applications. However, improving system performance under changing operating conditions remains a persistent engineering challenge, particularly for compact configurations where environmental and operational factors interact in a nonlinear manner.

Recent studies have demonstrated the growing role of optimization and predictive modelling approaches in improving renewable energy system performance. Moradzadeh et al. [1] employed Response Surface Methodology (RSM) to optimize hybrid energy systems through photovoltaic sizing while accounting for environmental and economic conditions and demonstrated its capability for supporting energy management decisions. Velasquez et al. [2] modelled the power coefficient and performance of Savonius wind turbines operating at low wind speeds are greatly impacted by the geometric changes. However, there are still few thorough optimization studies that incorporate blade twist angle and aspect ratio using factorial design and RSM, underscoring the necessity of methodical research

into these crucial design parameters. In the wind energy field, Shen et al. [3] combined computational fluid dynamics (CFD) with response surface analysis to improve the aerodynamic and operational characteristics of double Darrieus vertical-axis wind turbines. Keykhah et al. [4] further demonstrated the usefulness of RSM in optimizing fuel-cell-assisted wind energy systems across different operating environments and showed considerable improvements in energy output and carbon reduction potential.

Several investigations have focused specifically on improving turbine-level performance through geometric and operational optimization. Sanaye and Farvizi [5] applied RSM and CFD analysis to optimize helical blade configurations and demonstrated improved power coefficient prediction accuracy. Demie et al. [6] integrated CFD and RSM to enhance diffuser-augmented wind turbine performance through aerodynamic optimization of diffuser geometry. Cabrera-Escobar et al. [7] employed CFD to investigate the aerodynamic performance of Savonius vertical-axis wind turbines under various operating conditions. Numerical simulations have demonstrated the effectiveness of CFD in predicting torque, power coefficient, and flow characteristics for enhancing turbine performance and design. Jafaryar et al. [8] investigated asymmetric blade configurations using CFD and confirmed the importance of rotational speed in turbine performance. Mansoubi et al. [9] used wind tunnel experiments and CFD simulations to assess the impact of wind loads on wind turbine structural performance and stability. Few studies have been conducted to improve turbine geometry to obtain maximum aerodynamic performance and power coefficient through statistical design methods. In addition, Taghinezhad et al. [10] studied the optimized dual-rotor ducted wind turbines through statistical experimental design and demonstrated significant effects of rotor arrangement and flow conditions on power generation.

Chaudhuri et al. [11] noted the significance of WECSs in a sustainable energy supply and discussed the recent advances in turbine design, control strategies and maximum power-point tracking to achieve maximum energy extraction from wind. Modern wind turbines have been made more efficient and reliable through an ongoing advancement in materials for blades, generator technologies and power management systems. Ghoneam et al. [12] studied the integration of finite element analysis with optimization methods has also been explored for composite turbine structures, where aspect ratio and stacking sequence were identified as dominant design variables. Mahmoud and Oyediji [13] pointed out the success of adaptive and Model Predictive Control (MPC) strategies in enhancing the performance, energy capture, and operational reliability of wind turbines. The above control strategies offer efficient structures to improve the dynamic response and overall efficiency of WECSs. Alqahtani [14] investigated traffic-driven vertical-axis wind turbines and reported that installation location strongly influenced energy harvesting performance. In view of their compactness and versatility, the authors Wang et. al. [15] pointed out the increasing relevance of small wind turbines (SWTs) for sustainable energy harvesting and Internet of Things (IoT) applications. Recent developments have been centered around making SWT more efficient with innovative power generating mechanisms, optimized power generating systems, and more efficient energy harvesting technologies. To study the performance of wind turbine structures under different wind conditions, Huo and Tong [16] used finite element modelling and aerodynamic load analysis to investigate the dynamic response of wind turbine structures under wind loads. Their application to the safety, reliability, and structural design of wind turbines have been shown through experimental testing and numerical simulations to be effective.

Beyond geometric optimization, statistical experimental methodologies have increasingly been adopted for analysing system response under interacting operating conditions. RSM has emerged as a robust framework for evaluating nonlinear relationships among multiple variables while reducing experimental burden. Combined with Central Composite Design (CCD), the method supports predictive model construction and identification of favourable operating regions in engineering systems. Mohammed and Naik [17] further extended response-surface-based optimization to small wind turbine composite blade design and showed that multi-variable optimization can improve structural and operational performance simultaneously.

Although substantial progress has been achieved in turbine optimization and renewable energy system design, most existing studies primarily focus on large-scale configurations, geometric modification, or simulation-driven performance enhancement. Experimental investigations into compact tubular WECSs remain comparatively limited. In particular, the combined influence of operating duration and wind conditions on electrical response behaviour has not been systematically investigated using statistically guided modelling frameworks. Existing approaches also provide limited understanding of how interacting operating variables influence response characteristics and predictive reliability under constrained experimental conditions.

To address these limitations, this study investigates the experimental optimization and predictive modelling of a tubular WECS using response surface methodology based on Central Composite Design (RSM-CCD). The study evaluates the combined effects of operating time and wind speed on electrical current generation and establishes a predictive relationship using polynomial response modelling and analysis of variance (ANOVA)-based assessment. The objective is to identify favourable operating conditions while improving understanding of system response behaviour in compact wind energy applications. The proposed framework provides an experimentally supported methodology for performance-oriented analysis and contributes to the broader development of data-driven

optimization strategies for renewable energy systems.

2 Research Gap and Novelty of Study

Although considerable progress has been achieved in the optimization and performance improvement of renewable energy conversion systems, existing studies have predominantly focused on large-scale turbine configurations, geometric modification strategies, and simulation-driven aerodynamic analysis. Experimental investigations addressing compact WECSs operating under confined or tubular configurations remain comparatively limited. In particular, the response behaviour of such systems under varying operating conditions has not been systematically characterised through experimentally supported predictive frameworks.

Previous studies have demonstrated the effectiveness of optimization approaches such as CFD, statistical design methods, and response surface techniques in improving turbine efficiency and operational performance. However, most existing work has emphasized structural optimization or energy output enhancement without explicitly examining how interacting operating variables jointly influence electrical response characteristics in compact wind energy systems. In addition, conventional trial-and-error experimentation generally requires extensive testing effort while providing limited capability to quantify interaction effects and establish predictive relationships.

To address these limitations, this study adopts a statistically guided experimental optimization framework based on RSM and CCD for a tubular WECS. Rather than relying solely on isolated parameter evaluation, the proposed approach investigates the combined influence of operating time and wind speed on electrical current generation and establishes a predictive response model supported by ANOVA.

The novelty of this work lies in three aspects. First, an experimentally driven investigation is conducted on a compact tubular wind energy configuration that remains relatively underexplored compared with conventional turbine systems. Second, the study integrates response surface modelling with systematic operating-condition evaluation to reveal interaction effects governing electrical response behaviour. Third, a predictive framework is developed to identify favourable operating conditions while reducing experimental burden and improving interpretability of system performance.

The proposed methodology contributes to performance-oriented analysis of compact renewable energy systems and provides a reproducible framework for experimental optimization and predictive assessment in wind energy applications.

3 Methodology

A compact tubular WECS was developed for experimental evaluation of electrical response under controlled operating conditions. The system consisted of a fully enclosed wind turbine integrated with a generator, battery storage unit, inverter, and charge controller. The generator had a maximum rated power capacity of 1 kW and was connected to a maintenance-free battery pack for energy storage and output stabilization. The enclosed tubular arrangement was adopted to provide a compact operating environment and maintain stable turbine operation under fluctuating airflow conditions.

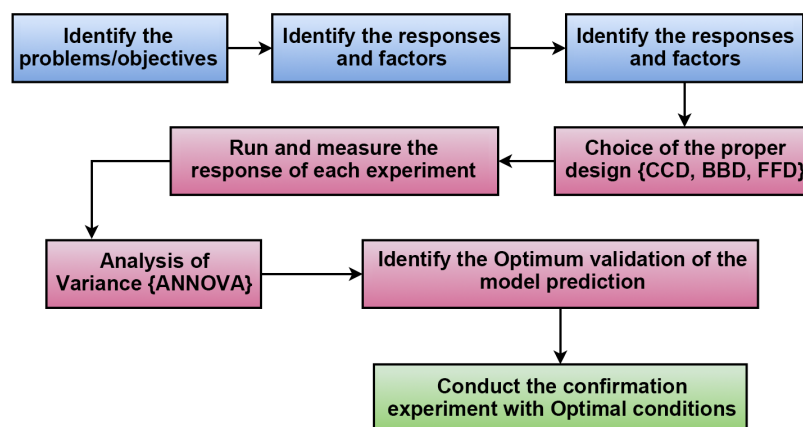


Figure 1. Flow chart of steps involved in response surface methodology based on central composite design (RSM-CCD) method [18]

Experimental data acquisition was conducted over a period of seven days. The wind energy system was installed on a roadside platform to capture airflow variations generated under practical outdoor conditions, including the influence of surrounding vehicular movement. Measurements were collected under different operating durations

ranging from 30–180 min. Electrical response was quantified using current output as the dependent variable, while operating time and wind speed were considered as independent variables for subsequent modelling and optimization.

To establish the relationship between operating conditions and electrical response, a predictive modelling framework based on RSM–CCD was implemented using Design-Expert software. The adopted framework enabled systematic evaluation of factor interactions while reducing the number of required experimental runs compared with conventional trial-and-error procedures. The overall workflow of the RSM–CCD approach used in this study is illustrated in Figure 1.

The total number of experimental runs was determined according to the standard RSM–CCD formulation shown in Eq. (1):

$$N = 2^k + 2k + n_c \quad (1)$$

where, N represents the total number of experimental runs, k denotes the number of independent variables, and n_c refers to the number of centre points.

A second-order polynomial response model was adopted to predict the dependent variable and quantify the interaction effects among operating parameters, as expressed in Eq. (2) [19]:

$$Y = \beta_0 + \sum_{i=1}^k \beta_i X_i + \sum_{i=1}^k \beta_{ii} X_i^2 + \sum_{i < j}^k \beta_{ij} X_i X_j + \varepsilon \quad (2)$$

where, Y denotes the predicted response; β_0 is the intercept; β_i is the linear coefficient; β_{ii} is the quadratic coefficient; and β_{ij} is the interaction coefficient. In addition, X_i and X_j are coded independent variables; k represents the number of independent variables; and ε represents the residual/error term. This modelling framework was employed to describe response behaviour and identify operating conditions associated with improved electrical performance.

4 Results and Discussion

Figure 2 illustrates the scatter and box plot among three variables time (min), wind speed (m/s) and current (Amp). The diagonal includes box plots of each of the variables, displaying their distributions, median values, interquartile ranges and outliers. The off-diagonal cells contain scatter plots that are used to show the correlation between two variables. As an example, the time versus wind speed plot can indicate any relationship between these two values, whereas the time versus current and wind speed versus current plots can evaluate the dependencies between these two values. These scatter plots are used in determining patterns where we have either a positive or a negative relationship and possible outliers that could be the indication of anomalies. In general, this matrix will give a holistic perspective of the interactions between these variables, which will help investigate deeper and continue to research or experiment in the related disciplines.

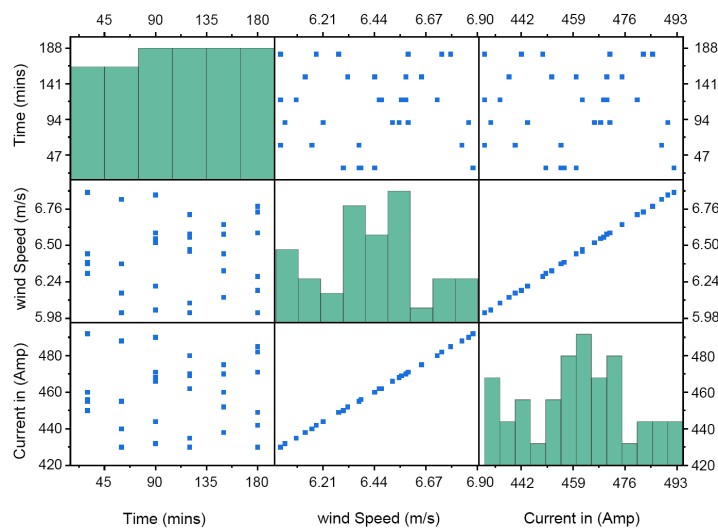


Figure 2. Scatter plot for the variables

4.1 Normal Distribution Plot for the Input and Output Response

Figure 3 shows the histograms with overlaid smooth density curves provide insight into how the experimental observations are distributed for each variable, which is important for interpreting model coverage and prediction reliability. From the time plot, the counts appear higher in the mid-range approximately from 60–120 min and lower near the extremes around from 20 to 40 and 160 to 180 min, indicating the dataset has stronger representation in central time conditions and weaker coverage at the tails. Similarly, the wind speed (m/s) distribution is centered roughly around from 6.4 to 6.6 m/s with fewer runs at both the lower end near 6 to 6.2 m/s and higher end at 6.7 to 6.8 m/s, suggesting moderate variability but most experimental information concentrates around typical wind-speed operating levels. For the current (Amp) response, the histogram shows that current values predominantly cluster around an intermediate band approximately from 455 to 470 Amp, with relatively fewer observations at lower 430 to 440 Amp and upper from 480 to 490 Amp ranges.

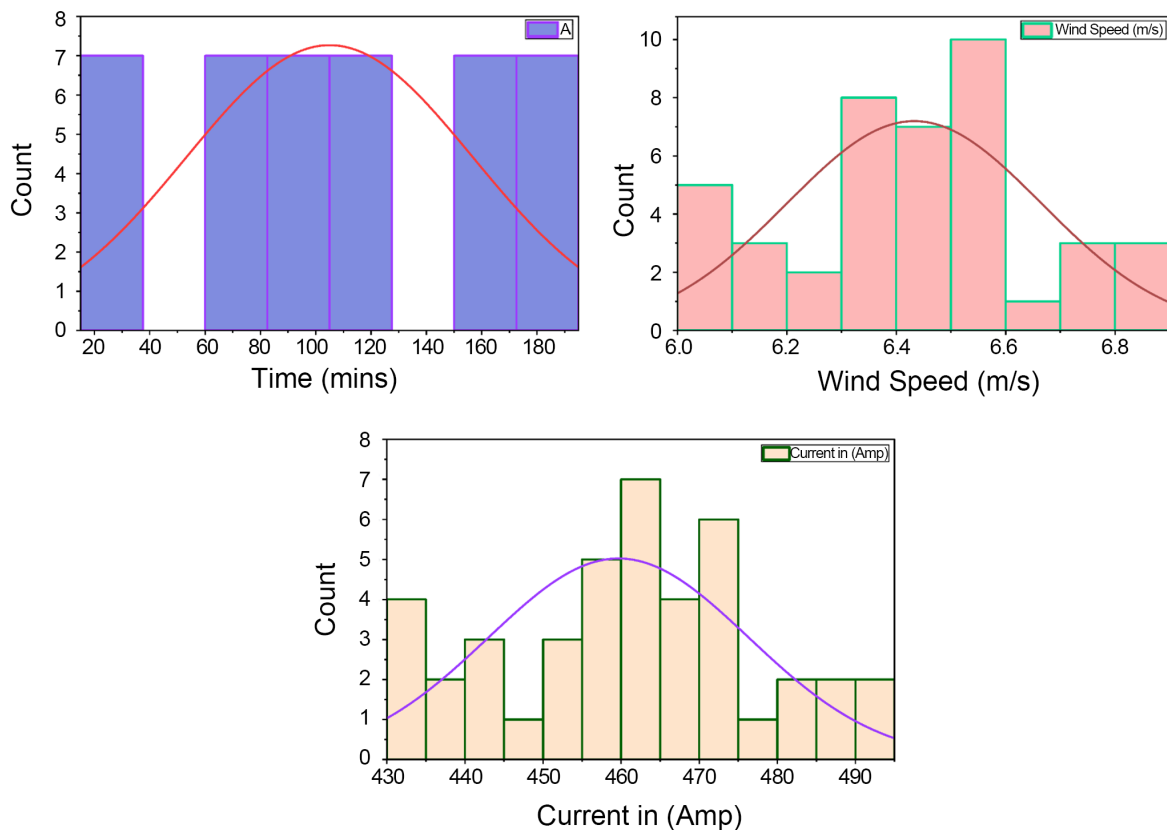


Figure 3. Displays the normal distribution for the response and variables

4.2 Response Surface Methodology Based on Central Composite Design Model Information and Analysis of Variance Statistical Analysis

Table 1 shows the 2-input factor used such as A: time in min and B: wind speed m/s are applied in a statistical analysis. Factor A: time is a numerical continuous variable with a minimum of 30 and a maximum of 180, coded as -1 to 30 when the value is lower and +1 to 180 when the value is higher. Such coding aids in the interpretation and making comparisons of findings in a standard range. Factor B: wind speed is between 6.0 and 6.88 m/s, with coded values of -1 to 6.0 indicating low values and +1 to 6.88 indicating high values. This framework can be used to better appreciate the way in which changes in the two factors affect the response variable in the analysis. The characteristics of the two factors imply that they are required to analyze their impacts on the outcomes of interest in the study, and this may imply their importance in additional statistical modeling [20].

Table 2 illustrates the ANOVA statistical analysis of input factors on the response variable which are represented in the source table of variation. This general model is a strong cause of the variability, with a sum of squares of 4818.81 and F -value 35.49 which, a p -value of less than 0.0001, gives a good level of significance. Analyzing the impact of individual factors, the linear effect of B (i.e., wind speed) is significant with $F = 139.92$, $p < 0.0001$. In

contrast, the linear effect of factor A (i.e., time) is not statistically significant with $F = 3.25$ and $p = 0.1146$. Their interaction, i.e. AB , is also non-significant ($F = 0.1473$, $p = 0.7125$). However, the squared terms of both A^2 and B^2 have a significant effect with $p = 0.0141$. The model's unexplained variation is 190.11 and insignificant lack of fit (i.e., $p = 0.2810$ is greater than 0.05) indicates that there is minor discrepancy between the model's prediction and actual experimental data due to random experimental noise. Thus, model fits the experimental data well.

Table 1. Factor information details for the response surface methodology (RSM) design

Factor	Type	Minimum	Maximum	Coded Low	Actual Low	Coded High	Actual High
A: Time (min)	Numeric	30	180	-1	30	+1	180
B: Wind Speed (m/s)	Numeric	6.0	6.88	-1	6.0	+1	6.88

Table 2. Analysis of variance (ANOVA) statistical analysis for the response

Source	Sum of Squares	Degrees of Freedom	Mean Squares	F -value	p -value	
Model	4818.81	5	963.76	35.49	<0.0001	significant
A: Time	88.17	1	88.17	3.25	0.1146	
B: Wind Speed	3800.17	1	3800.17	139.92	<0.0001	
AB	4.00	1	4.00	0.1473	0.7125	
A^2	286.77	1	286.77	10.56	0.0141	
B^2	286.77	1	286.77	10.56	0.0141	
Residual	190.11	7	27.16			not significant
Lack of Fit	110.11	3	36.70	1.84	0.2810	
Pure Error	80.00	4	20.00			
Cor Total	5008.92	12				

Using coded factors in Eq. (3) provides an opportunity to make predictions regarding the response at levels of each factor and the high levels are coded as +1 and low levels are coded as -1. The coded equation can be found especially helpful in the evaluation of the relative impacts of the factors by analyzing the coefficients related to each of them. Conversely, the Eq. (4) in real terms is usable in predicting with the original units of the factors. Nonetheless, it is not to be applied in the comparison of the relative impacts of the factors because the coefficients are scaled to show the units of each variable, and the intercept is not placed at the middle of the design space.

Equations in terms of coded factors:

$$\text{Current (Amp)} = 443.52 - 3.83A + 25.17B + 1.00AB + 10.19A^2 + 10.19B^2 \quad (3)$$

Equations in terms of actual factors:

$$\begin{aligned} \text{Current (Amp)} = & 2303.85799 - 0.626676(\text{Time}) - 623.89166(\text{Wind Speed}) \\ & + 0.030303(\text{Time} \times \text{Wind Speed}) + 0.001811(\text{Time}^2) + 52.63252(\text{Wind Speed}^2) \end{aligned} \quad (4)$$

Figure 4a shows the normal probability distribution of residuals in the error terms of the fitted regression model, which is approximately normalized. A critical assumption for the validity of an ANOVA as most residuals follow this straight reference line, especially in the middle. Slight deviations at the tails are noticeable, but it is expected that there are at least a few observations whose residual magnitudes are somewhat higher or lower than normal. But these are not massive, they are just a few that aren't bad enough. The near linearity reinforces the belief that the model residuals meet the normality assumption well enough to be used in predictive and inferential models. The residuals of the regression have the characteristics expected of the residuals from a linear model.

The Predicted vs. Actual plot in Figure 4b is used to evaluate the predictive accuracy of the response by comparing predicted response values with observed (actual) response values. The data points found to be close to 45° with ideal behavior line. It demonstrates the model produces predictions which are generally consistent with measured outcomes. The points are spread relatively evenly around the diagonal which suggests low prediction error over the range of response points. The relationship between the predictors and the response variable is well represented by

the model. There is no strong systematic curvature or clustering far from the diagonal, suggesting that the model has no major biases like under predicting in one region and over predicting in another. There is good overall agreement between the predicted and actual data in the fitted model.

Figure 5a represents the residual variance vs. predicted value plot which is used to determine if there is constant homoscedasticity and if there are systematic lack-of-fit patterns in the model. Residuals should be randomly distributed around the zero line without a definite trend or change in the spread of the residuals as values get larger. As per the plots, the residual points are concentrated around the point with zero, and the distribution does not exhibit a definite funnel shape or exhibit a strong nonlinearity over the range of the prediction. This indicates that there is a relatively stable variance at the fitted values, thus supporting the assumption of homoscedasticity. Modelling structures are not suggested if there is any small scatter about zero. Thus, the residual behavior confirms that the model is correctly specified and that the assumptions that are necessary for stable model estimation and valid uncertainty quantification are fairly met.

The residuals versus run plot as per Figure 5b can help to detect problems such as drift over time, changes in experimental conditions that were not controlled, and non-random grouping of runs because of the order of runs executed. If an experiment is done under ideal conditions, the residuals should not have any periodic trend or systematic trend up or down with the number of the runs, but the residuals should be randomly distributed around the value of zero. The plot shows the plot of residuals with no obvious repeating pattern or evidence of run order effects or temporal instability. A few have slight positive or negative residuals, but there is no obvious pattern of residuals in the order of the runs.

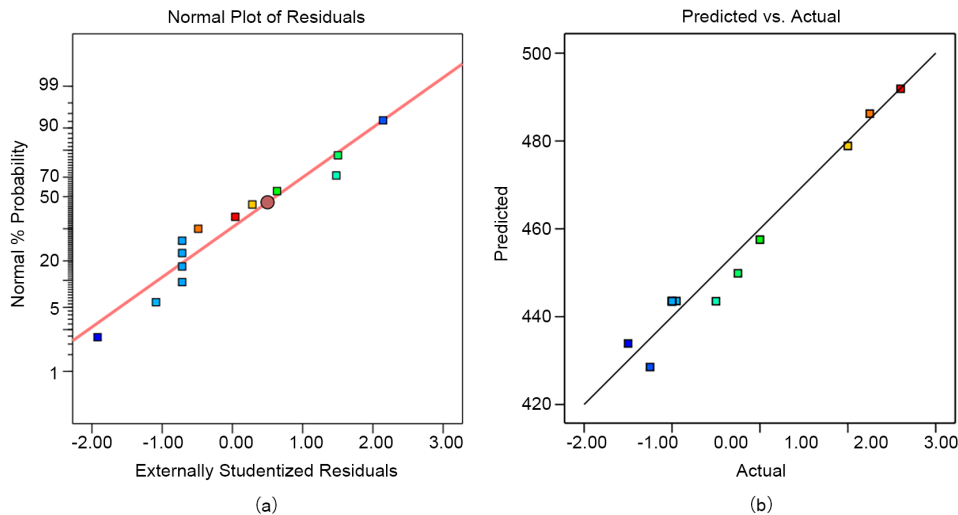


Figure 4. Normal of residuals and actual vs. predicted value

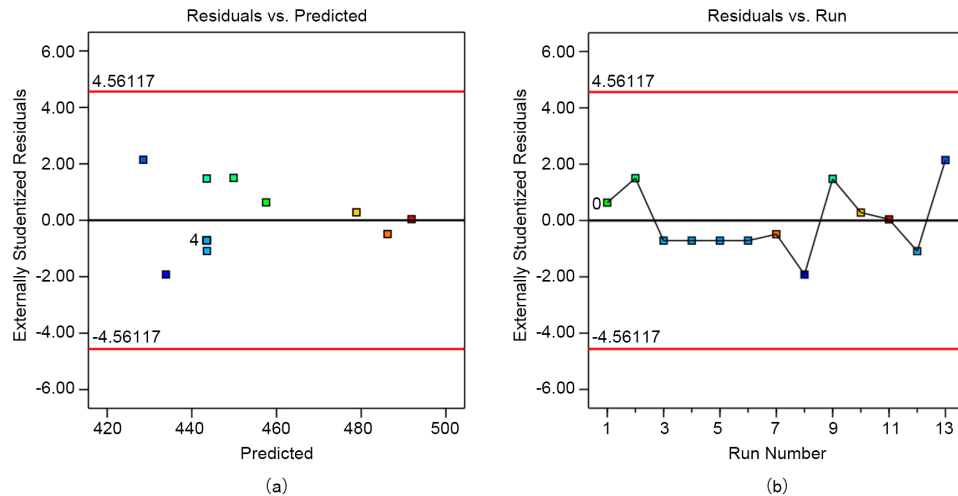


Figure 5. Residuals vs predicted, and residuals vs. run plots

Figure 6a illustrates the difference in fits (DFFITS) plot is used to detect influential observations, which are the ones that significantly influence the values of the fitted model. A large absolute DFFITS means that there is a potential outlier with a large influence on the regression fit and may cause a change in the coefficients and a decrease in the robustness of the model. Most of the DFFITS values are in the control region between the reference thresholds plotted, where there is no single observation largely dominating the fitted model. Some runs have larger DFFIT's values, but the sizes do not seem to be too large and are generally within the influence acceptable range. This shows that the regression estimates are not too dependent on any individual data point, leading to more confidence in the stability of the model.

The difference in beta estimates (DFBetas) plot in Figure 6b assesses the influence that each observation has on the estimated regression intercept and can be used to identify observations that may lead to changes in the baseline prediction level. When DFBetas values are relatively small and do not exceed the limits plotted within the graph, it means that no observation significantly affects the estimate of intercept. The value of DFBetas in the presented figure is mostly within the range of (0, 0.2) indicating that the intercept parameter is not significantly affected by most observations. Several points are slightly off, but not so much that they allude to big excursions outside the critical bounds, which implies that the estimation for the intercept is steady. This stability is crucial because if there is a large intercept influence it may be caused by either model misspecification, an outlier, or improper experimental conditions for some of the runs.

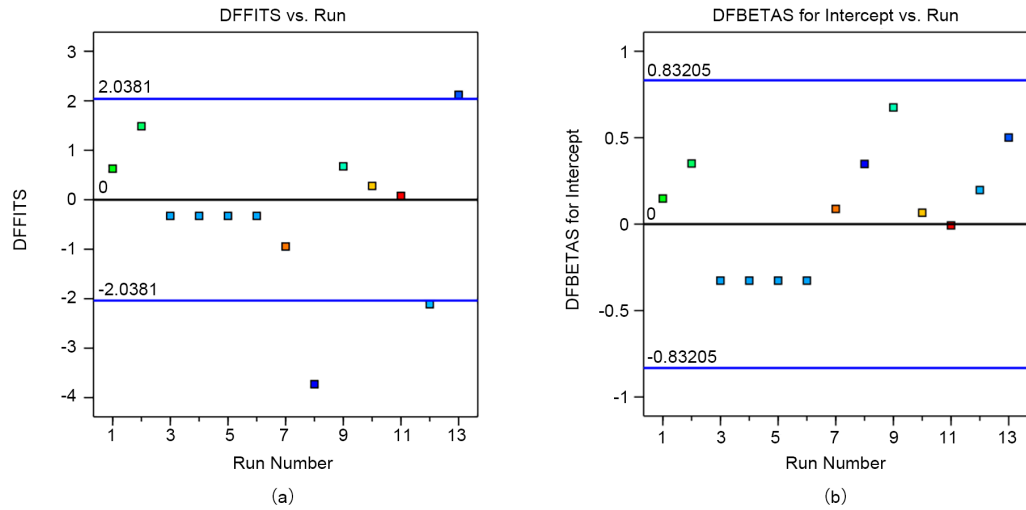


Figure 6. Difference in fits (DFFITS) and difference in beta estimates

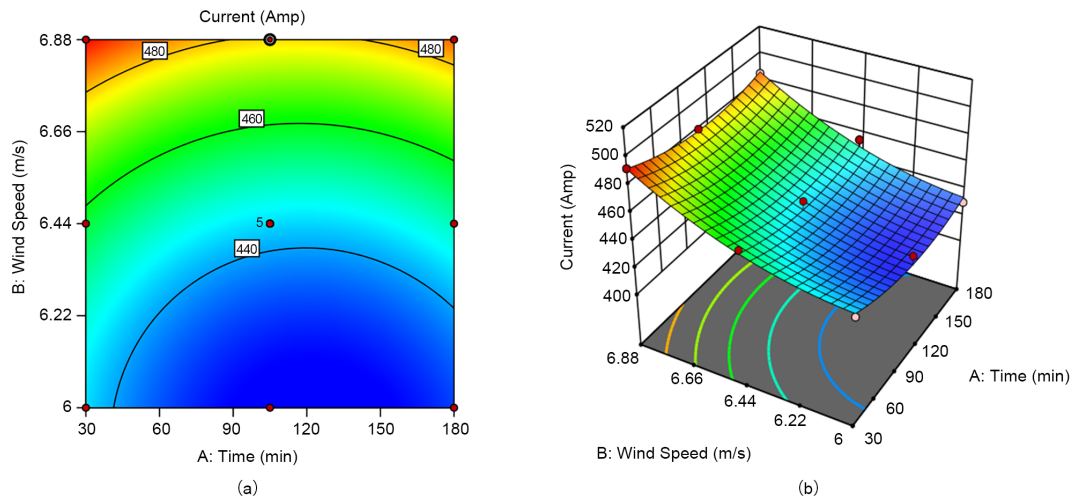


Figure 7. 2D contour and 3D surface plots

Figure 7a shows the 2D contour map of the response variable Current (Amp) with respect to Wind Speed (B) and Time (A). It shows a smooth transition in color and a symmetrical curved line indicating a non-linear relationship

between input factors and output current. The peak current can be seen at peak wind speed. The contour lines are parallel with curvature geometry without any distortion. This behavior aligned well with the findings of the ANOVA which confirmed that individually both factors shaped the profile and mutual interaction is non-significant statistically ($p = 0.7125$).

The model obtained can be geometrically interpreted using the 3D response surface plot as displayed in Figure 7b, which depicts the combined effect of Time (A) and Wind Speed (B) on Current (Amp). The surface is not flat but curved, which means that there are nonlinear terms in the model, or that there are quadratic effects or large interactions between factors. The shape indicates that current has relatively higher values in some regions and lower values in others, thus creating a ridge or valley-type of behavior. The predicted points are also plotted and are generally in the same relative position around the surface, which is consistent with the fitted model being a good representation of the experimental data. Optimization point of view, the surface shows the direction of fastest change in current with operating conditions and can be used to find regions of interest for optimization without trial-and-error.

The developed model has high predictive power and reliability. The model had a low standard deviation of 5.21, indicating the experimental and predicted values had a minimum variation, and the mean response value of the model was 452.92 Amp. The coefficient of variation (C.V.%) was 1.15%, which is relatively low and indicates high precision and reproducibility of the experimental results. The coefficient of determination $R^2 = 0.9620$ demonstrates that 96.20% of the variability in the response can be explained by the model. In the same way, the adjusted R^2 value of 0.9349 demonstrates good agreement between predicted and experimental values. The predicted R^2 value is 0.8089, which yields a difference of 0.1260 between the Adjusted and Predicted R^2 score. The difference below 0.2 indicates that the model is not overfitted, with excellent prediction performance.

Additionally, the Adequate Precision value of 17.89 is quite high and well above the desired value of 4, showing that there is an adequate signal to noise ratio. This means that the developed model is sufficiently robust and can indeed be used to navigate and optimize the design space.

The regression plot shown in Figure 8 compares the actual and predictive values of a response factor. The plot has points that are the forecasted values of Current (in Amperes) as compared to the measured values. A dashed line is used to show the optimal fit, which is the case when the actual values are equal to the predicted values. The inset statistics have indicated that the predictive and the actual value are strongly correlated, with a value of R^2 equal to 0.9620. These measurements indicate that the prediction model is good as the forecasts are very close to the real values and show that the model is useful in modeling.

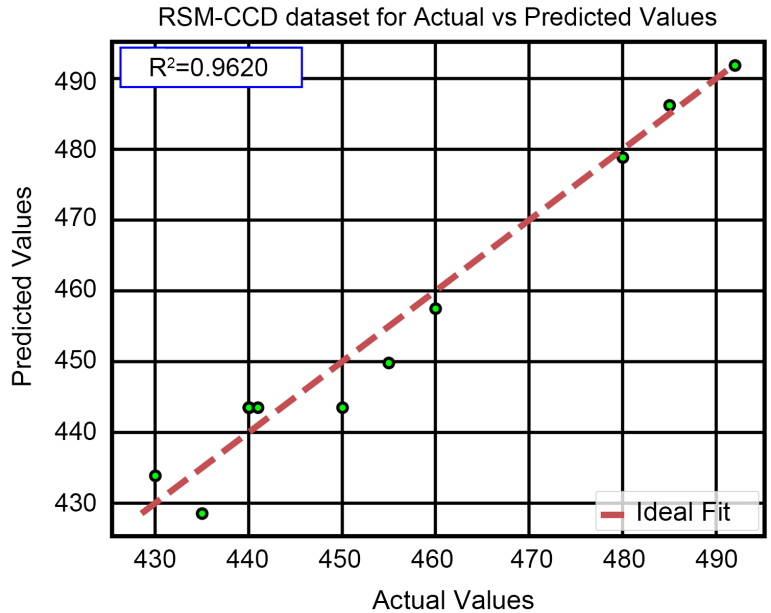


Figure 8. Regression plot for the response variable

Figure 9 shows a graph that is an optimization of the output variable. Factor A is denoted on the left panel with time of 30 minutes and at the top limit. Factor B is displayed as the middle panel with wind speed being 6.88 m/s which is also marked by a red dot at the maximum limit. The equivalent Current value in the lower panel is about 491.897 Ampere which is represented on the vertical axis. Besides, the general desirability of such a solution is observed as 0.998, indicating a good result following the selected values of the variables. The visualization provides an effective summary of the optimum settings that can be used to reach the desired Current output.

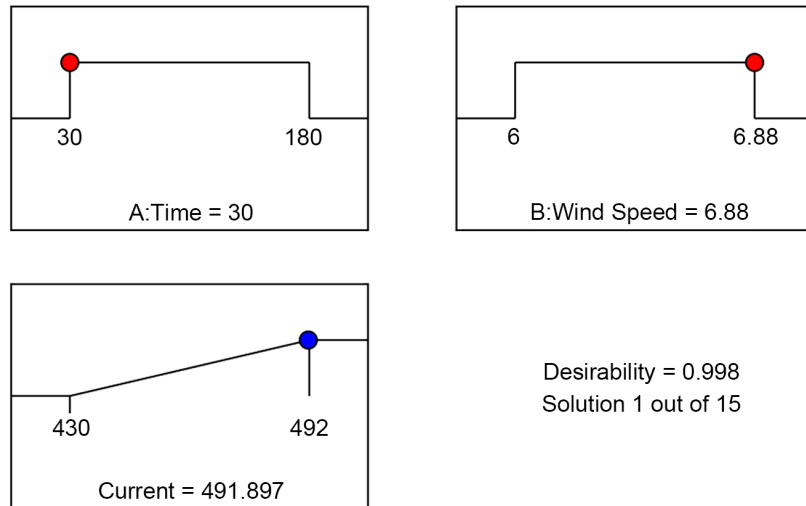


Figure 9. Ramp function plot for the response

Note: RSM-CCD: response surface methodology based on central composite design.

Figure 10 shows the relationship between time, wind speed and current is shown in the correlation heatmap. A perfect positive correlation $r = 0.99$ is seen between wind speed and current, which means that as wind speed increases, the current increases directly and proportionately. This implies that in the system, wind speed has a greater impact on the current generation. Time exhibits a very small positive correlation $r = 0.026$ with wind speed and $r = 0.030$ with current, meaning that for the range of these parameters observed, time has negligible impact on both wind speed and current.

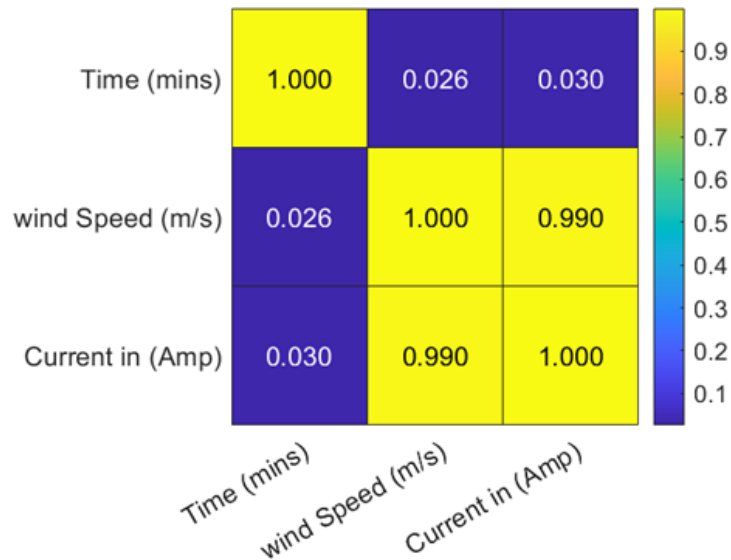


Figure 10. Correlation matrix for the input and output variables

5 Conclusion

This study demonstrated the applicability of RSM-CCD for the experimental optimization and predictive modelling of a tubular WECS under varying operating conditions. The proposed framework enabled systematic evaluation of the combined effects of operating time and wind speed on electrical current generation and supported the development of a statistically reliable response model. The regression analysis confirmed strong agreement between predicted and experimental observations, with a coefficient of determination (R^2) of 0.962, indicating that the developed model effectively captured the response behaviour of the system within the investigated operating range. Among the examined variables, wind speed showed the dominant influence on electrical output, while

interaction and nonlinear effects contributed to the overall response characteristics.

The developed quadratic model provided a statistically significant representation of the experimental data (Model: $F = 35.49$, $p < 0.0001$), while the non-significant lack-of-fit ($F = 1.84$, $p = 0.2810$) confirmed the adequacy of the proposed model for describing the system response. The ANOVA results revealed that wind speed was the most influential factor affecting current generation ($F = 139.92$, $p < 0.0001$), whereas the linear effect of operating time was not statistically significant ($F = 3.25$, $p = 0.1146$). However, the quadratic terms associated with operating time (A^2) and wind speed (B^2) were statistically significant ($F = 10.56$, $p = 0.0141$), indicating the presence of nonlinear response behaviour. The model's unexplained variation is 190.11 and insignificant lack of fit (i.e., $p = 0.2810$ is greater than 0.05) indicates that the model fits the experimental data well even in the presence of some random noise.

Optimization results identified favourable operating conditions at a time duration of 30 min and a wind speed of 6.88 m/s, under which the system achieved a current output of approximately 491.897 A with a desirability value of 0.998. These findings demonstrate that the adopted modelling approach can support efficient identification of operating conditions without extensive experimental iterations. Overall, the study provides an experimentally supported framework for analysing system response and optimizing compact wind energy systems. The proposed approach contributes to the broader application of data-driven experimental optimization methods and offers a reproducible methodology for performance-oriented assessment of renewable energy conversion systems.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The author declares that generative AI or AI-assisted technologies were used only for language editing during manuscript preparation. All scientific content, interpretations, and conclusions were developed and verified by the author, who takes full responsibility for the manuscript.

References

- [1] A. Moradzadeh, K. Pourhossein, A. Ghorbanzadeh, M. Nazari-Heris, I. Colak, and S. M. Muyeen, "Optimal sizing and operation of a hybrid energy systems via response surface methodology (RSM)," *Sci. Rep.*, vol. 14, no. 1, p. 20226, 2024. <https://doi.org/10.1038/s41598-024-71035-9>
- [2] L. Velasquez, J. Rengifo, Saldarriaga, A. Rubio-Clemente, and E. Chica, "Geometric optimization of Savonius vertical-axis wind turbines using full factorial design and response surface methodology," *Sci.*, vol. 7, no. 4, p. 154, 2025. <https://doi.org/10.3390/sci7040154>
- [3] Z. Shen, S. G. Gong, H. X. Zu, and W. Y. Guo, "Multi-objective optimization study on the performance of double Darrieus hybrid vertical axis wind turbine based on DOE-RSM and MOPSO-MODM," *Energy*, vol. 299, p. 131406, 2024. <https://doi.org/10.1016/j.energy.2024.131406>
- [4] A. Keykhah, E. Assareh, S. Mobayen, S. Hoseinzadeh, F. Mancini, J. K. Alagarasan, and D. A. Garcia, "Feasibility assessment and response surface optimization of a fuel cell-integrated sustainable wind farm in Italy," *Int. J. Hydrogen Energy*, vol. 179, p. 151536, 2025. <https://doi.org/10.1016/j.ijhydene.2025.151536>
- [5] S. Sanaye and A. Farvizi, "3-D CFD and response surface method for optimizing helical blade vertical axis wind turbines," *Appl. Eng. Sci.*, vol. 26, p. 100318, 2026. <https://doi.org/10.1016/j.apples.2026.100318>
- [6] A. B. Demie, V. R. Ancha, and M. B. Kahsay, "Enhancing diffuser augmented wind turbine performance: Aerodynamic optimization of diffuser's cross-section via CFD and response surface methodology," *Results Eng.*, vol. 30, p. 110539, 2026. <https://doi.org/10.1016/j.rineng.2026.110539>
- [7] J. Cabrera-Escobar, C. M. Carrillo Rosero, C. H. Arroba Arroba, S. P. Cabrera Anda, C. Cabrera-Escobar, and R. Cabrera-Escobar, "CFD simulation of a vertical-axis Savonius-type micro wind turbine using meteorological data from an educational environment," *Clean Technol.*, vol. 8, no. 2, p. 40, 2026. <https://doi.org/10.3390/clean-technol8020040>
- [8] M. Jafaryar, R. Kamrani, M. Gorji-Bandpy, M. Hatami, and D. D. Ganji, "Numerical optimization of the asymmetric blades mounted on a vertical axis cross-flow wind turbine," *Int. Commun. Heat Mass Transfer*, vol. 70, pp. 93–104, 2016. <https://doi.org/10.1016/j.icheatmasstransfer.2015.12.003>
- [9] S. Mansoubi, H. Sadeghi, Y. Ma, and R. Mohebbi, "An experimental and numerical investigation into the influence of wind effects on wind turbines with tubular towers," *Ocean Eng.*, vol. 313, p. 119555, 2024. <https://doi.org/10.1016/j.oceaneng.2024.119555>

- [10] J. Taghinezhad, R. Alimardani, M. Masdari, and E. Mahmoodi, "Performance optimization of a dual-rotor ducted wind turbine by using response surface method," *Energy Convers. Manag.* X, vol. 12, p. 100120, 2021. <https://doi.org/10.1016/j.ecmx.2021.100120>
- [11] A. Chaudhuri, R. Datta, M. P. Kumar, J. P. Davim, and S. Pramanik, "Energy conversion strategies for wind energy system: Electrical, mechanical and material aspects," *Materials*, vol. 15, no. 3, p. 1232, 2022. <https://doi.org/10.3390/ma15031232>
- [12] S. Ghoneam, A. Hamada, and T. Sherif, "Modeling and optimization for the dynamic performance of vertical-axis wind turbine composite blades," *J. Sol. Energy Eng.*, vol. 143, no. 2, p. 021005, 2021. <https://doi.org/10.1115/1.4048159>
- [13] M. S. Mahmoud and M. O. Oyedele, "Adaptive and predictive control strategies for wind turbine systems: A survey," *IEEE/CAA J. Autom. Sin.*, vol. 6, no. 2, pp. 364–378, 2019. <https://doi.org/10.1109/JAS.2019.1911375>
- [14] M. Alqahtani, "Optimization of VAWT installation with spatial and temporal complexity considerations," *Front. Energy Res.*, vol. 12, p. 1373586, 2024. <https://doi.org/10.3389/fenrg.2024.1373586>
- [15] H. Wang, B. Xiong, Z. Zhang, H. Zhang, and A. Azam, "Small wind turbines and their potential for internet of things applications," *iScience*, vol. 26, no. 9, p. 107674, 2023. <https://doi.org/10.1016/j.isci.2023.107674>
- [16] T. Huo and L. Tong, "Wind-induced response analysis of wind turbine tubular towers with consideration of rotating effect of blades," *Adv. Struct. Eng.*, vol. 23, no. 2, pp. 289–306, 2020. <https://doi.org/10.1177/1369433219865815>
- [17] S. Mohammed and R. L. Naik, "Multi-objective optimization of glass/carbon hybrid composites for small wind turbine blades using extreme mixture design response surface methodology," *Funct. Compos. Struct.*, vol. 6, no. 2, p. 025009, 2024. <https://doi.org/10.1088/2631-6331/ad45a7>
- [18] N. Shambira, G. Makaka, and P. Mukumba, "Velocity augmentation model for an empty concentrator-diffuser-augmented wind turbine and optimisation of geometrical parameters using surface response methodology," *Sustainability*, vol. 16, no. 4, p. 1707, 2024. <https://doi.org/10.3390/su16041707>
- [19] Z. Kaseb and H. Montazeri, "Data-driven optimization of building-integrated ducted openings for wind energy harvesting: Sensitivity analysis of metamodels," *Energy*, vol. 258, p. 124814, 2022. <https://doi.org/10.1016/j.energy.2022.124814>
- [20] S. L. Lee and S. Shin, "Wind turbine blade optimal design considering multi-parameters and response surface method," *Energies*, vol. 13, no. 7, p. 1639, 2020. <https://doi.org/10.3390/en13071639>